

Testing of Hypothesis: Examples

Reading Objectives:

In this reading, you will learn about different models related to testing of hypothesis concerning mean. If the hypothesis involves mean, then we consider a suitable model accordingly.

Main Reading Section:

1. Testing of Hypothesis: One Mean—Large Sample

Example: The mean of ball bearings manufactured by a company is 73.2 inches, with a standard deviation of 8.6 inches. If a sample of 45 is chosen at random and its mean is 76.7 inches, test the significant difference between the means as to which one is superior at a 1% level of significance.

Solution:

Null hypothesis $H_0: \mu = 73.2$.

Alternative hypothesis H_1 : There is a significant difference such that $\mu > 73.2$ (one-tailed test).

Level of significance = 0.01.

Since the sample is larger, the test statistic is

$$Z = (\bar{x} - \mu) / (\sigma / \sqrt{n}) \\ = (76.7 - 73.2) / (8.6 / \sqrt{45}) = (3.5 \times 6.7) / 8.6 = 2.72$$

Critical region: at a 1% level of significance, for a one-tailed test, the critical value is $z = 2.33$.

Since the calculated value of $z (=2.72)$ is greater than the tabulated value (2.33), the null hypothesis H_0 is rejected.

Decision: Thus, we find that there is a significant difference between the sample and population mean.

Note: If the population variance is unknown and the sample is large ($n \geq 30$), then we use z-test. In the test statistic, we replace the population variance with sample variance (s^2).

2. Testing of Hypothesis: One Mean—Small Sample

Example: The specifications for a certain kind of ribbon call for a mean breaking strength of 180 pounds. If five pieces of the ribbon (randomly selected from different rolls) have a mean breaking strength of 169.5 pounds with a standard deviation of 5.7 pounds, test the null hypothesis $\mu = 180$ pounds against the alternative hypothesis $\mu > 180$ pounds at the 0.01 level of significance. Assume that the population distribution is normal.

Solution:

Null hypothesis: $\mu = 180$ pounds

Alternative hypothesis: $\mu > 180$ pounds (one-sided test)

Level of significance: $\alpha = 0.01$

Criterion: 3.747 is the value of $t_{0.01}$ for $(5-1=)$ 4 degrees of freedom

Calculations:

The test statistics is $t = (\bar{x} - \mu) / (S / \sqrt{n}) = (169.5 - 180) / (5.7 / \sqrt{5}) = -4.12$

$|t| = 4.12$

Decision: Since $|t| = 4.12 > 3.747$, i.e., cal. value $|t| > \text{tab. } t$ ($t_{0.01}$ for 4 d.f. = 3.747), the null hypothesis must be rejected at level $\alpha = 0.01$. In other words, the breaking strength is below specifications.

3. Testing of Hypothesis: Two Means—Large Sample

Example: An experiment was conducted to compare two products, A and B, which follows normal with standard deviations of 4.0 and 5.0, respectively. A sample of size 20 has an average life of 29.8 months, whereas a sample of product B with sample size 25 has an average life of 34.7 months. Can we conclude that at a 0.01 level of significance, the average life of these products is different?

Solution:

Null hypothesis as $H_0: \mu_1 - \mu_2 = 0$

Alternate hypothesis $H_1: \mu_1 - \mu_2 \neq 0$

Level of significance: $\alpha = 0.01$

$$\text{Test statistic: } z = \frac{(\bar{x}_1 - \bar{x}_2) - d}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{29.8 - 34.7}{\sqrt{\frac{16}{20} + \frac{25}{25}}} = -3.66$$

Critical region is z in $[-2.58, 2.58]$ since it is a two-tailed test.

Decision: Calculated $z = -3.66$ lies in the region of rejection; hence reject H_0 .

4. Testing of Hypothesis: Two Means—Small Sample

Example: An experiment was conducted to compare two products, A and B. A sample of size 12 has an average life of 85 months with a standard deviation of 4 for product A. Similarly, a sample of product B with sample size 10 has an average life of 81 months with a standard deviation of 5. Can we conclude that at 0.05 level of significance, the average life of product A exceeds that of product B by 2. (Assume that the populations are approximately normal with equal variances.)

Solution:

Null hypothesis as $H_0 : \mu_1 - \mu_2 = 2$

Alternate hypothesis $H_1 : \mu_1 - \mu_2 > 2$

Level of significance: $\alpha = 0.05$

$$\text{Test Statistic: } t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} =$$

$$s_p = \sqrt{\frac{(11)(16) + (9)(25)}{12 + 10 - 2}} = 4.478,$$

$$t = \sqrt{\frac{(85 - 81) - 2}{4.478 \sqrt{\frac{1}{12} + \frac{1}{10}}}} = 1.04,$$

Critical region is $t > 1.725$ at a 5% level of significance with $12 + 10 - 2 = 20$ degrees of freedom.

Decision: Calculated $t = 1.04 < 1.725$ (table value); hence accept H_0 .